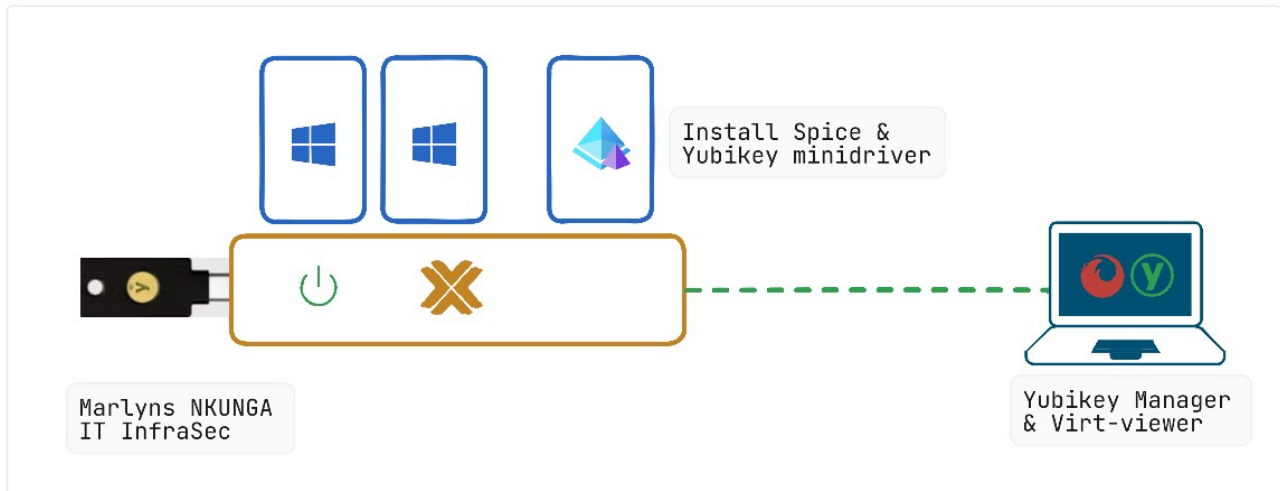


Moving beyond password to passwordless

Overview

The main goal is to configure Active Directory Passwordless Authentication with Yubikey, for the hardware constraint, I used Proxmox VE to complete this project, there are some requirements must be met on the client virtual machines. You'll find these tools in the tools folder.



Notice that, **Virt-viewer** and **Yubikey manager** must be installed on the management machine, Yubikey Manager to prepare the Yubikey, such as change the PIN, PUK and disable unused protocols.

First step : Prepare your virtual machines,

- Install OS, IP addresses configuration
- Promote a domain controller

Second step : requirements installation

These tools must be installed on your virtual machines, it's simple just next over and over.

- Spice Guest Tool
- Yubikey minidriver
- Yubikey Manager
- Virt-viewer
- Yubikey 5C NFC Serie

The YubiKey smartcard minidriver must be installed on all machines where the YubiKey will be used as a smartcard to access, these include servers which users remotely connect to.

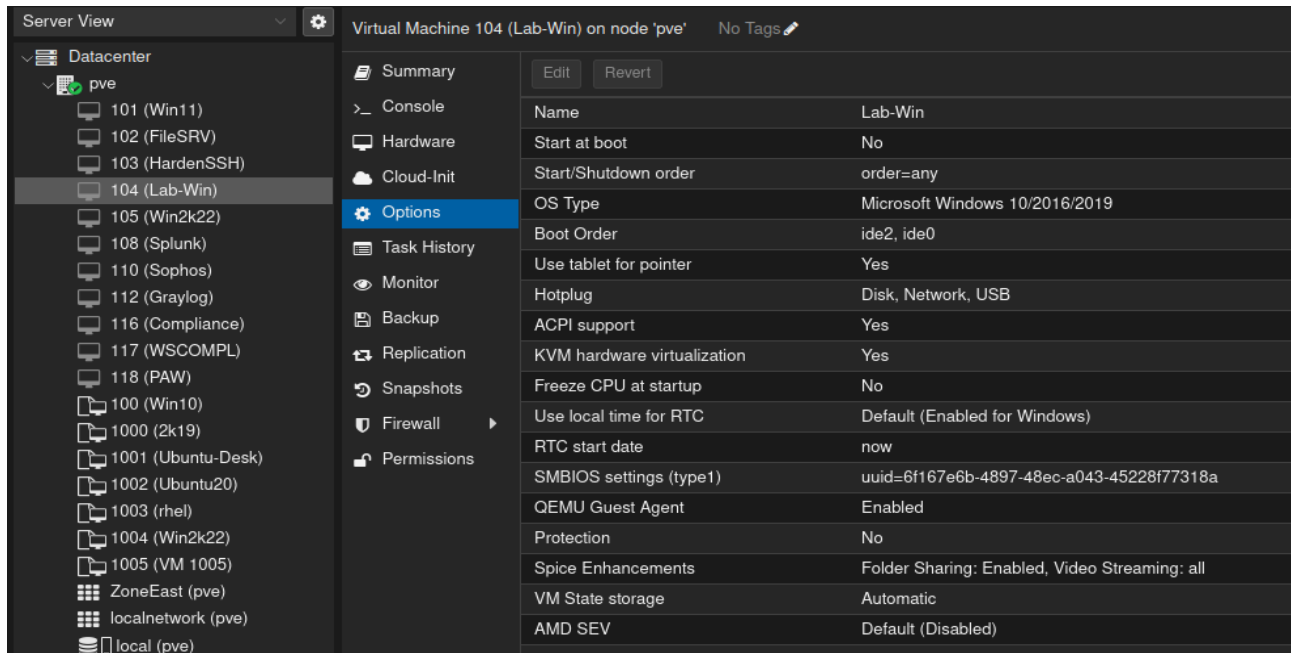
Notice that, When deploying the YubiKey Smartcard Minidriver to remote servers where the YubiKey cannot be physically inserted, a legacy node must be created to load the YubiKey Smartcard Minidriver. You'll find a PowerShell in tools folder, it helps you to install **Yubikey minidriver** in legacy mode on the servers.

msiexec /i YubiKey-Minidriver-latest-x64.msi INSTALL_LEGACY_NODE=1 /quiet

Third step : Spice Guest tool configuration on the virtual machines

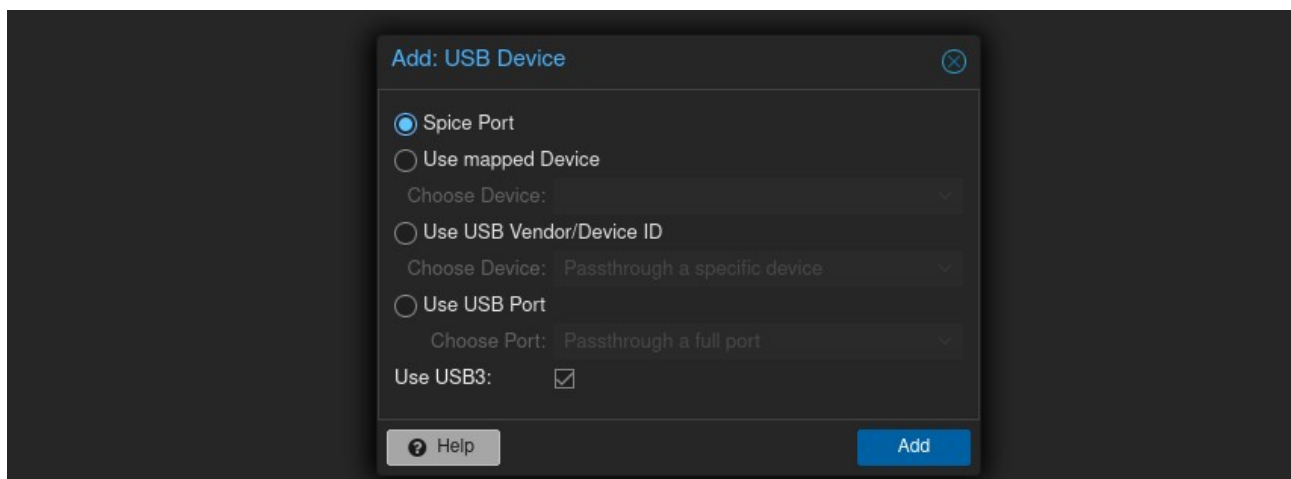
After configuring this tool the VM will be able to detect the Yubikey, when it'll be plugged.

Select the VM, Navigate to Options > Edit QEMU Guest Agent > select Use QEMU Guest Agent > Ok

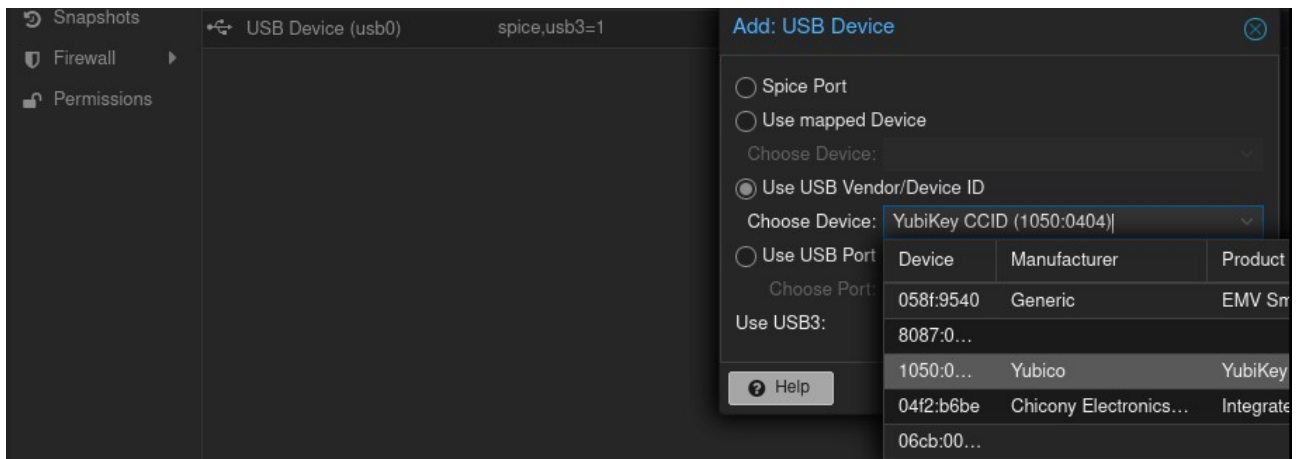


Notice that : QEMU Guest Agent must be enabled, as you see on the screenshot

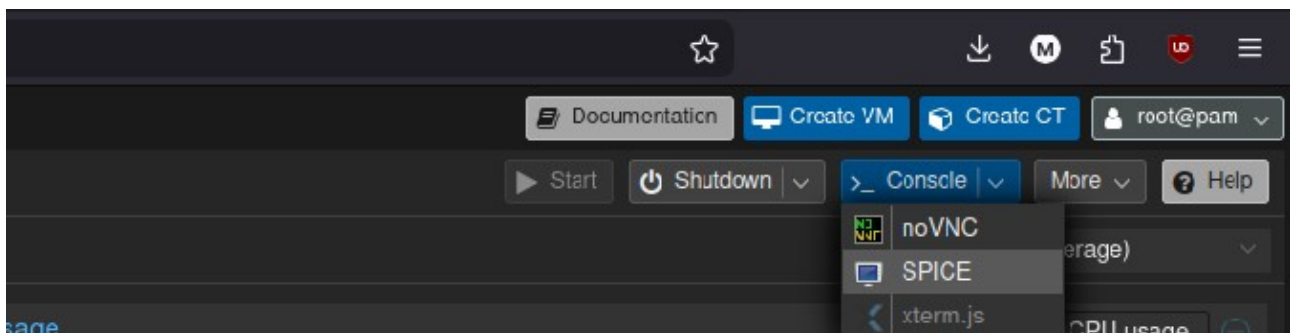
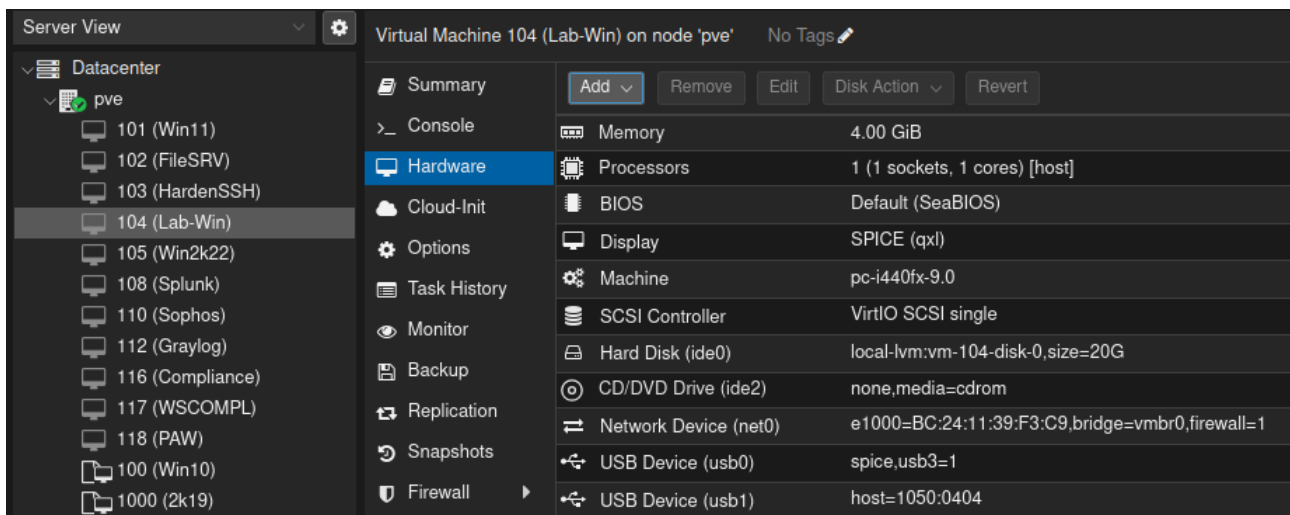
Select the VM > Navigate to Hardware > Add > USB Device > Select Spice Port and Use USB3 > Add



Select the VM > Navigate to Hardware > Add > USB Device > Select Use USB Vendor/Device ID > Yubico > Add



Final result

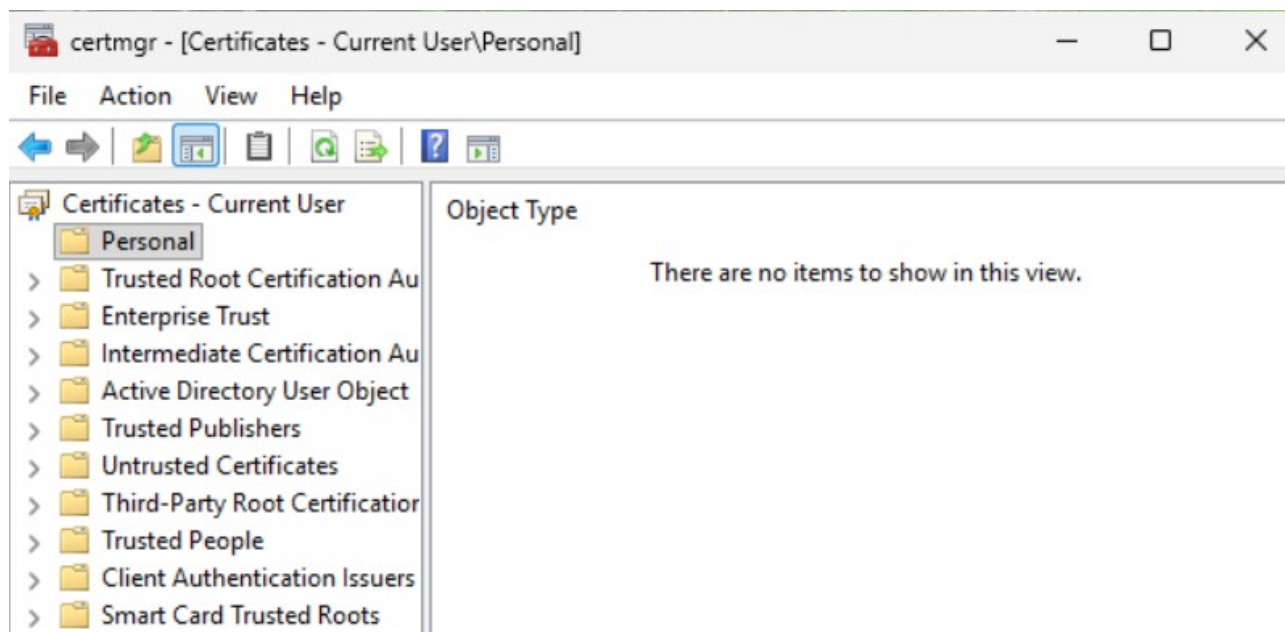


Notice that, Spice need the virt-viewer application on the management machine

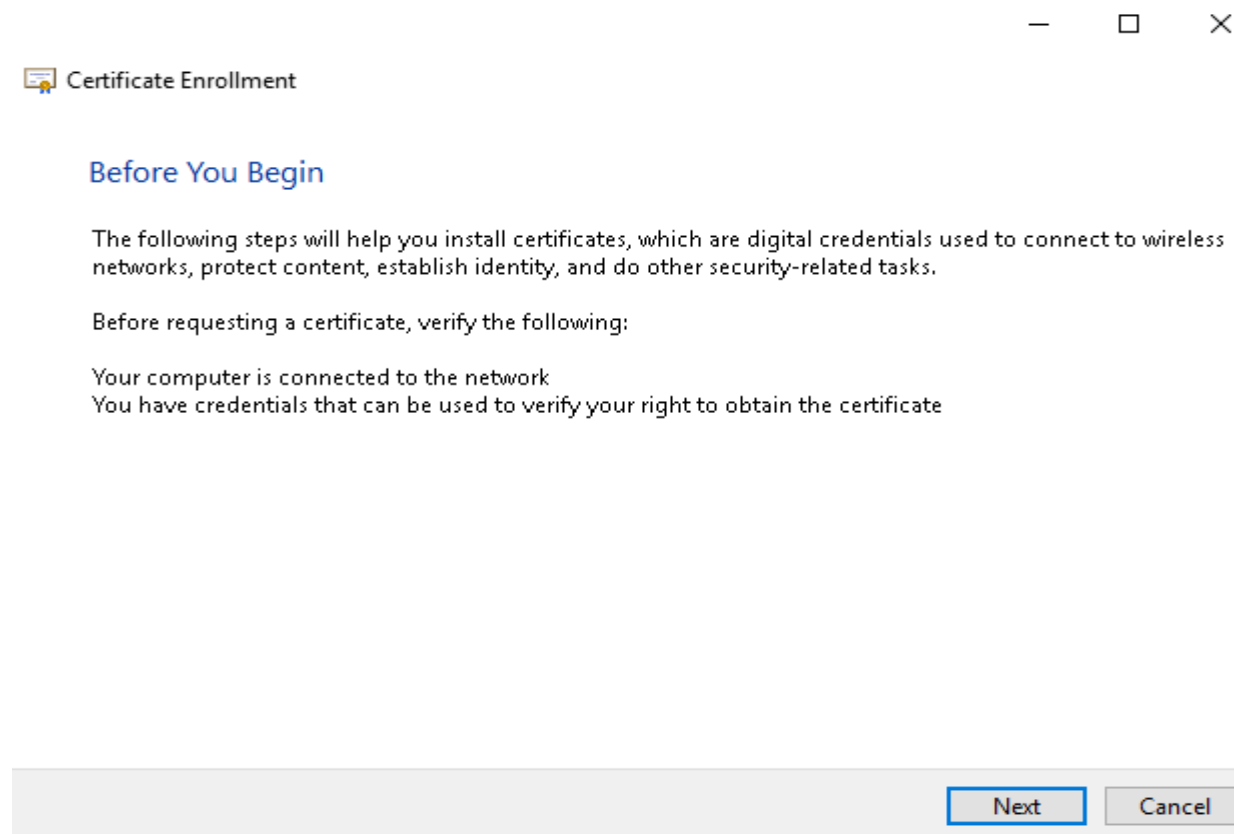
Fourth Step : Enrollment

Quickadcs script will create a security group called **Yubikey Authentication**, only the member of this group can enroll to Yubikey Smartcard logon certificate

Launch this command **certmgr.msc** in Execute



Navigate to Certificates > Personal Right Click > All task > New Request



 Certificate Enrollment

Select Certificate Enrollment Policy

Certificate enrollment policy enables enrollment for certificates based on predefined certificate templates. Certificate enrollment policy may already be configured for you.

Configured by your administrator

Active Directory Enrollment Policy 

Configured by you [Add New](#)

Next

Cancel

 Certificate Enrollment

Request Certificates

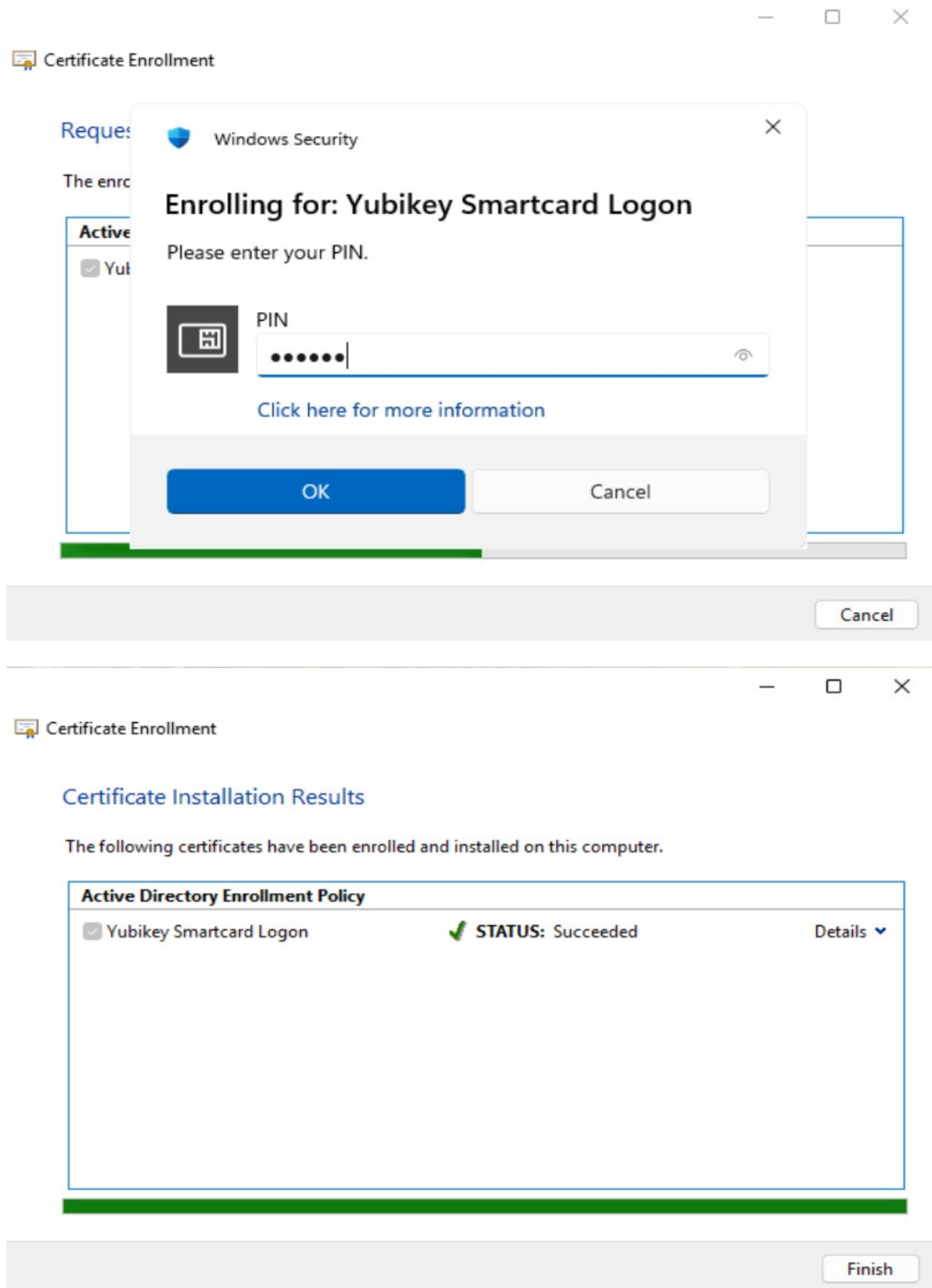
You can request the following types of certificates. Select the certificates you want to request, and then click Enroll.

Active Directory Enrollment Policy		
☐ Basic EFS	 STATUS: Available	Details 
☐ User	 STATUS: Available	Details 
☒ Yubikey Smartcard Logon	 STATUS: Available	Details 

☐ Show all templates

Enroll

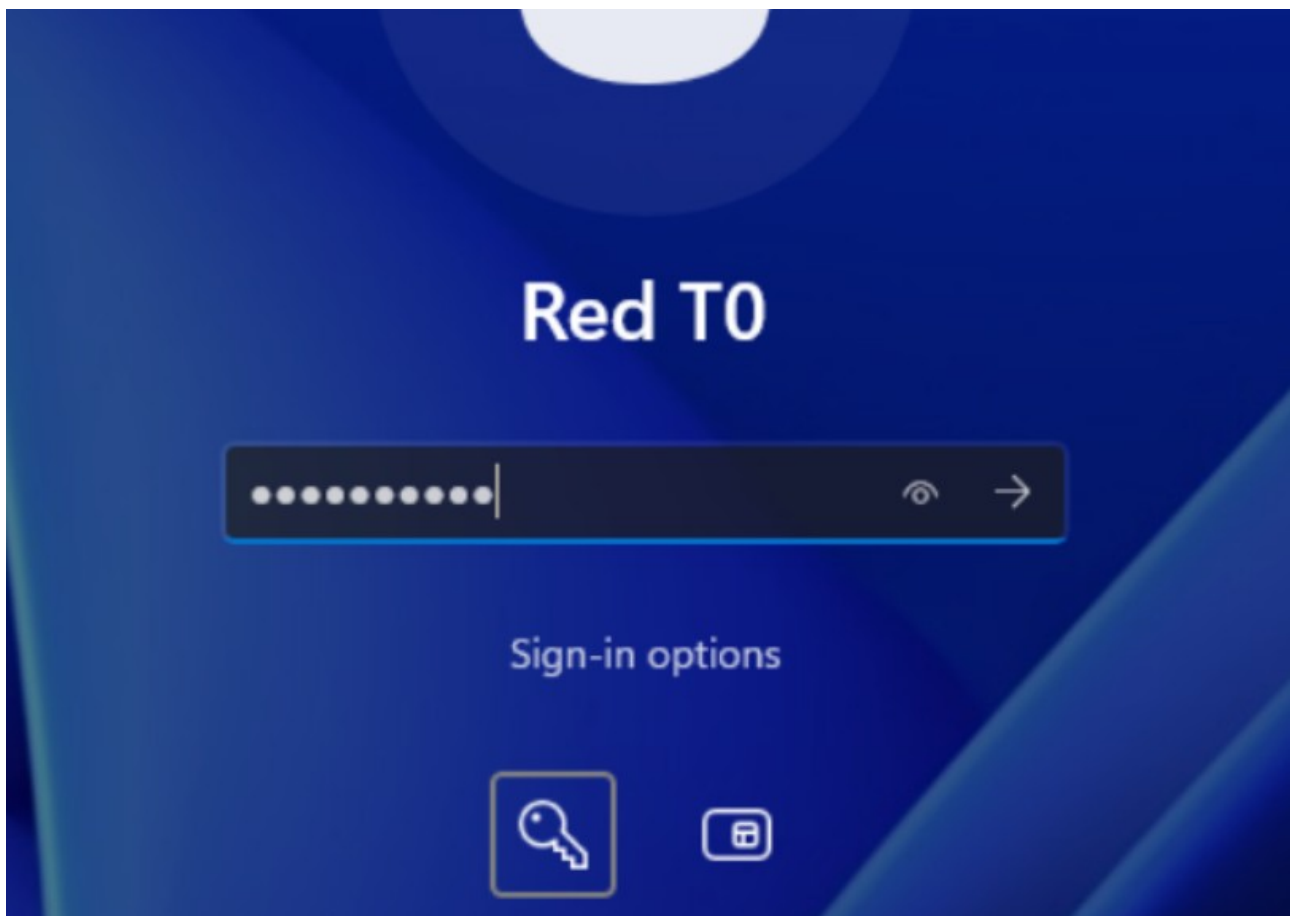
Cancel

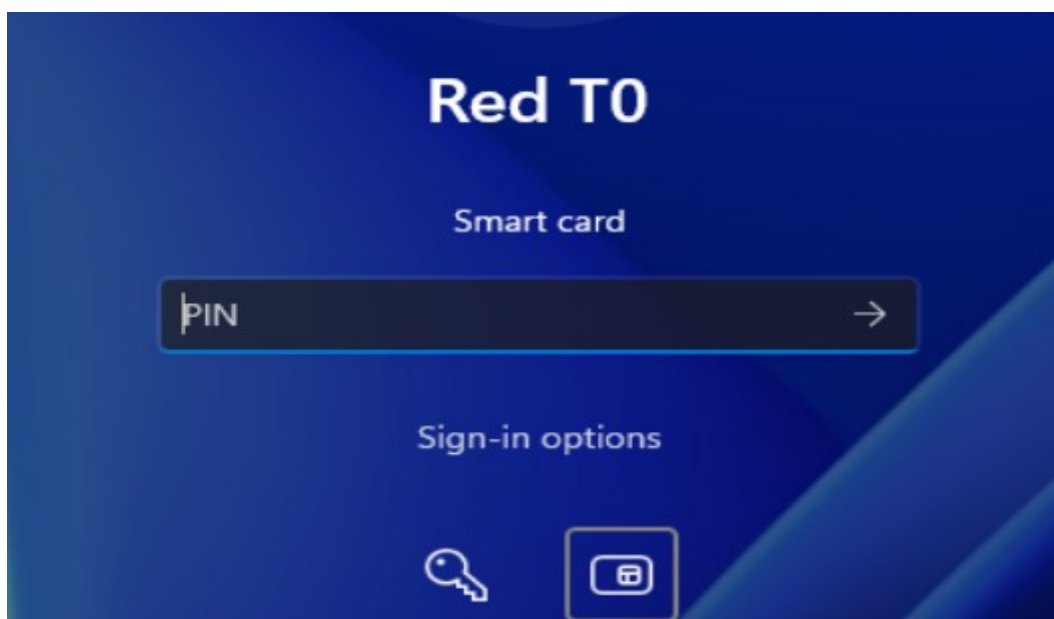
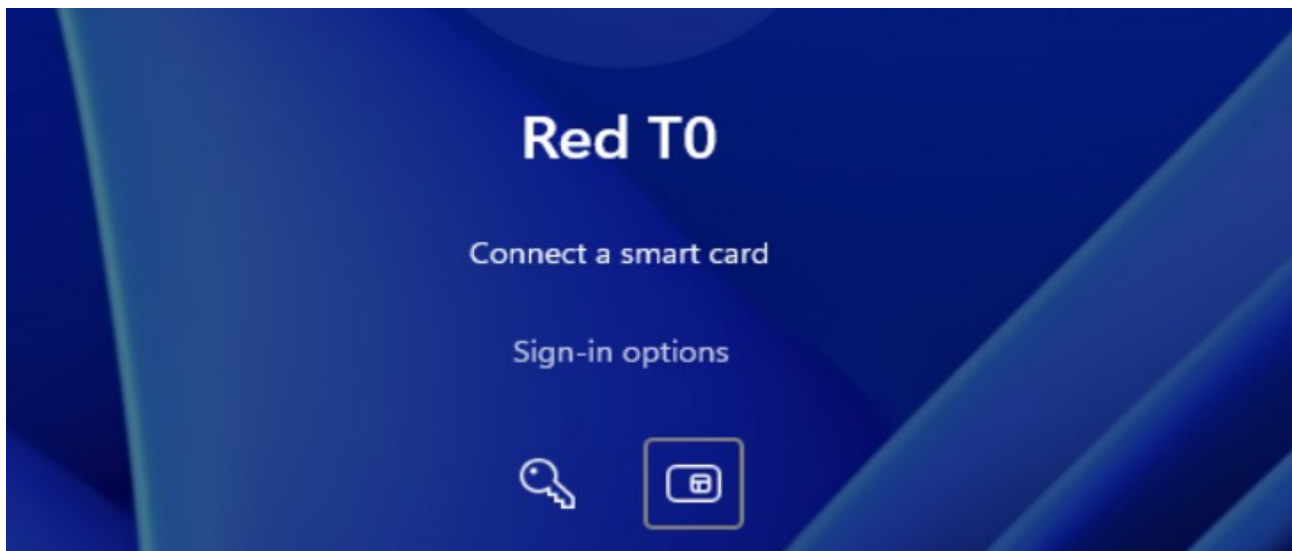


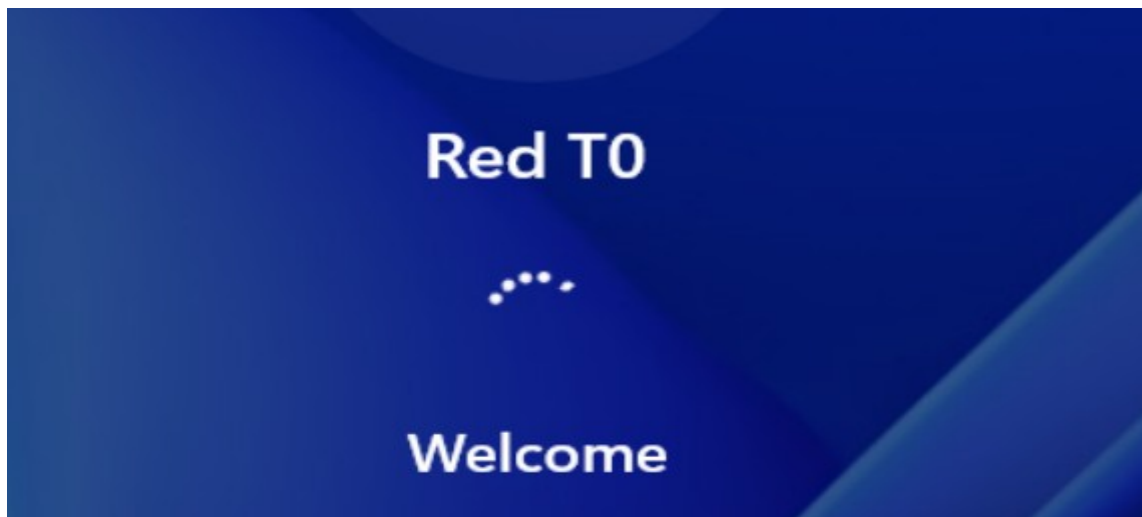
After enrollment, update the certificate in the Yubikey with this command, Run cmd **certutil -scroots update**

```
Command Prompt
Microsoft Windows [Version 10.0.26200.6584]
(c) Microsoft Corporation. All rights reserved.

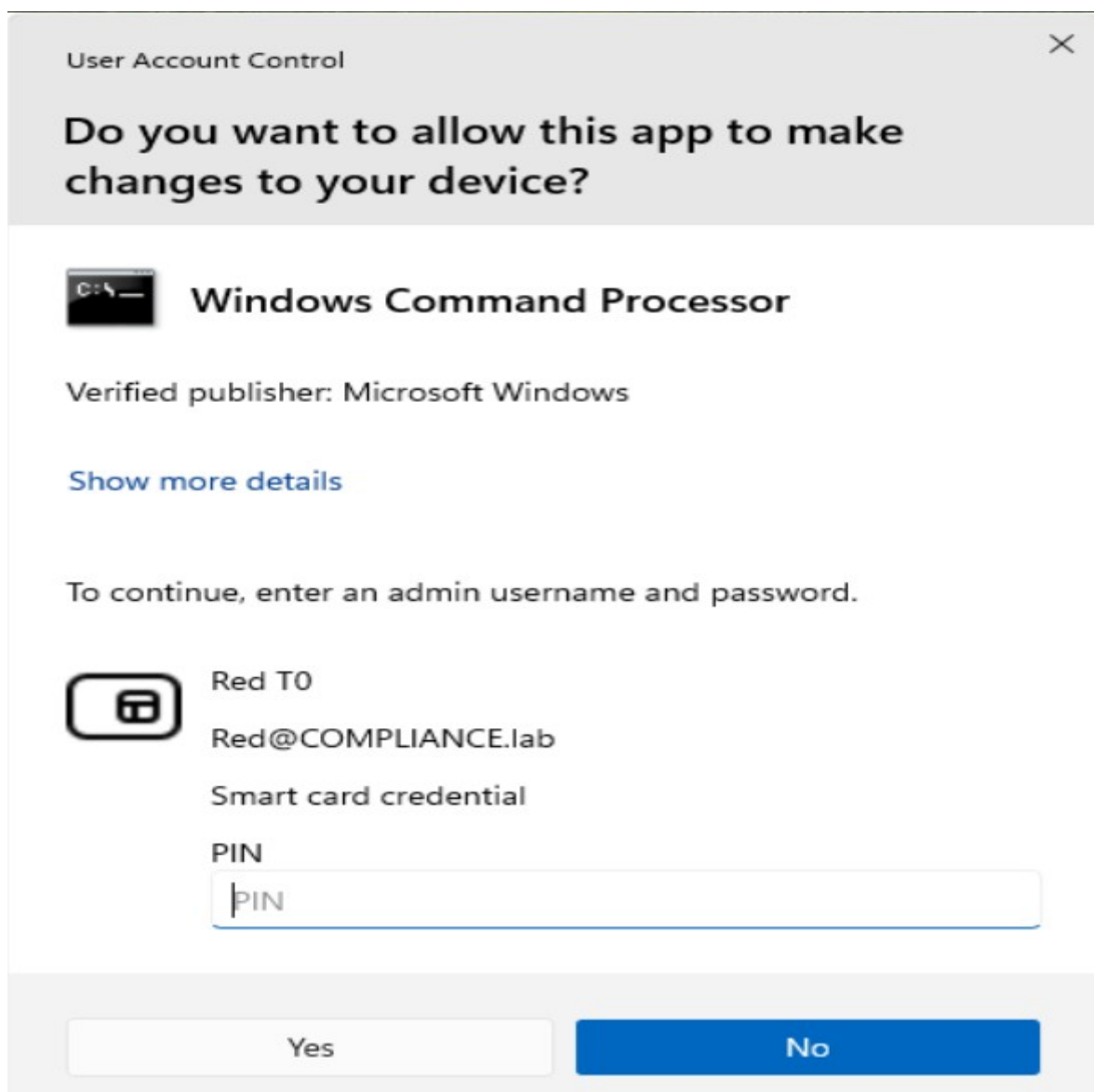
C:\Users\red>certutil -scroots update
Element 0:
Serial Number: 5ac2e5ee02bd918440040018c5b57f03
Issuer: CN=COMPLIANCE-ROOT-CA, DC=COMPLIANCE, DC=lab
```







Run CMD as Administrator



Remote Access using RDP

